

Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Beamwidth of the course array.** The ADALM-PHASER array has $N = 8$ patch elements on $d = 14$ mm centers and operates at 10.3 GHz. Use $\theta_{\text{HP}} \approx 0.886\lambda / (Nd \cos \theta_0)$ throughout.

- (a) Find the wavelength and the element spacing in wavelengths.

$$d/\lambda = \boxed{}$$

- (b) Compute the half-power beamwidth at broadside.

$$\theta_{\text{HP}} = \boxed{}$$

- (c) Compute the half-power beamwidth with the beam steered to $\theta_0 = 45^\circ$.

$$\theta_{\text{HP}} = \boxed{}$$

- (d) In one or two sentences, explain physically why the beam widens by $1/\cos \theta_0$ when the array is steered, even though no hardware changed.

2. **Nulls and the first-null beamwidth.** Same array as Question 1: $N = 8$, $d/\lambda = 0.481$, so $Nd = 3.85\lambda$. The nulls of a uniform array are at $\sin \theta = \sin \theta_0 \pm m\lambda / (Nd)$ for $m = 1, 2, 3, \dots$, excluding multiples of N .

(a) With the beam at broadside, find every null angle on the positive side of the pattern.

nulls =

(b) Compute the first-null beamwidth at broadside, and the ratio of it to the half-power beamwidth from Question 1(b).

FNBW =

(c) The array factor of an 8-element array has 7 nulls per period, but part (a) found only 3 on each side of broadside. Explain in one or two sentences where the others went.

(d) Steer the beam to $\theta_0 = 30^\circ$ and find the two nulls that bracket the main lobe, then the first-null beamwidth.

FNBW =

3. **Directivity of a uniform linear array.** Use $D \approx 2Nd/\lambda$ for a uniform broadside array of isotropic elements.

- (a) Compute the directivity of the 8-element course array, in linear units and in decibels.

$D =$

- (b) Suppose the same 8 elements were half-wave dipoles rather than patches. Estimate the total gain of the array in dBi.

$G_{\text{total}} =$

- (c) Turning off the outer four elements leaves the center four running. State what happens to the directivity and to the half-power beamwidth, with numbers, and explain in one sentence why both changes have the same cause.

4. **Sizing an array.** A ground-based X-band tracker at 10 GHz requires a half-power beamwidth of 2.0° and must steer to $\pm 60^\circ$.

- (a) Find the aperture length required at broadside, in millimeters and in wavelengths.

$$Nd =$$

- (b) The grating-lobe criterion is $d < \lambda / (1 + |\sin \theta_{0|\max}|)$. Find the largest usable spacing for this scan volume, then the element count if you build the array at $d = \lambda/2$.

$d_{\max} =$

$N =$

- (c) A colleague proposes $d = 0.9\lambda$ instead, which reaches the same 25.4λ aperture with 28 elements rather than 51 – a large saving in phase shifters and amplifiers. Evaluate the proposal: compute where the grating lobe lands at the edge of the scan volume, and state whether you would accept it.

- (d) The beamwidth requirement must hold across the whole scan volume, not just at broadside. State the aperture this actually demands and the resulting element count at $d = \lambda/2$.

5. **Reading a measured sweep.** A student sweeps the course array at 10.3 GHz ($\lambda = 29.1$ mm) with the GUI and records a main lobe 13.1° wide at half power, a null-to-null width of 28.5° , and a first sidelobe at -11.5 dBc.

- (a) From the measured half-power width alone, infer the aperture in millimeters and the number of active elements on 14 mm centers.

$N =$

- (b) Theory puts the first-null beamwidth at 30.1° and the student measured 28.5° . The sweep steps the commanded angle by the phase-shifter resolution of 2.8125° . Explain the discrepancy in one or two sentences.

- (c) The measured first sidelobe is -11.5 dBc, while theory for a uniform 8-element array gives -12.8 dBc. Is this evidence of a fault? Answer in one or two sentences.

- (d) A second sweep, taken after a lab partner changed the element gains, shows a main lobe 27° wide with no visible sidelobes above the noise floor. State which aperture configuration this is and what it implies about the number of active elements.

$N =$

Documentation: _____